Online Supplementary Appendix

The Readiness Option: Conservation Priority over Space and Time

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This supplement gives the derivations and proofs for the results in the main paper. Sections, equations, and tables numbered with the prefix S are in this supplement; all other references are to the main paper.

S.1 The option value at the later conflict and the readiness value

Let $dX_t = \mu X_t\,dt + \sigma X_t\,dW_t$, with $r > \max\{\mu, 0\}$ and $\sigma > 0$, and write $\mathcal{L}f(x) = \frac{1}{2}\sigma^2 x^2 f''(x) + \mu x f'(x)$ for its generator. The acquisition-window closure and conflict-arrival times are independent exponential variables with rates $\gamma_R > 0$ and $\lambda > 0$, also independent of W. Direct benefit $B \geq 0$ and acquisition cost $P > 0$ are fixed. Pre-existing capacity S is maintained; acquisition adds $\kappa > 0$, of which $\kappa e^{-\delta t}$ remains after t periods, where $\delta \geq 0$. Capacity when the conflict arrives fixes its access and intervention-cost terms thereafter. Thus acquiring after that arrival cannot improve the terms of that conflict.

Assume p and I are twice continuously differentiable on the capacity intervals considered, with $0 \leq p \leq 1$, $p' > 0$, $I > 0$ and $I' < 0$. Let $\beta > 1$ solve

$$\tfrac{1}{2}\sigma^2\beta(\beta-1) + \mu\beta - r = 0, \qquad z(u) = \frac{\beta I(u)}{\beta - 1}.$$

The standard perpetual investment option (**?**, Chapter 5) and its access-adjusted counterpart are

$$F(x; I(u)) = \begin{cases} \dfrac{I(u)}{\beta-1}(x/z(u))^{\beta}, & x < z(u), \\ x - I(u), & x \geq z(u), \end{cases} \qquad \mathcal{F}(x,u) = p(u)F(x; I(u)).$$

The readiness value is

$$K(x, S, \kappa) = \mathbb{E}_x\Big[e^{-rT_C}\{\mathcal{F}(X_{T_C}, S + \kappa e^{-\delta T_C}) - \mathcal{F}(X_{T_C}, S)\}\Big]. \tag{S.1}$$

Put $g(x,u) = \mathcal{F}_S(x,u)$. Integrating the capacity derivative and changing variables gives the identity

$$K(x, S, \kappa) = \int_0^{\kappa} ds \int_0^{\infty} \lambda e^{-(r+\lambda+\delta)t}\mathbb{E}_x[g(X_t, S + se^{-\delta t})]\,dt. \tag{S.2}$$

This identity accommodates any finite increment and any nonnegative depreciation rate.

S.2 Adaptive acquisition: proof of Proposition ?? and comparative statics

Set $d = r + \gamma_R + \lambda$ and let $\alpha > \beta$ solve

$$\tfrac{1}{2}\sigma^2\alpha(\alpha-1) + \mu\alpha - d = 0.$$

Suppressing fixed arguments of K, define

$$Q(x) = K(x) - \frac{xK_x(x)}{\alpha}. \tag{S.3}$$

Lemma S.2.1 (Adaptive acquisition)**.** *If $L = P - B > 0$, the acquisition problem*

$$W(x) = \sup_\tau \mathbb{E}_x[e^{-d\tau}\{K(X_\tau) - L\}]$$

includes non-acquisition at value zero and has a unique acquisition threshold $a > 0$, determined by $Q(a) = L$. Its value is

$$W(x) = \begin{cases} [K(a) - L](x/a)^\alpha, & x < a, \\ K(x) - L, & x \geq a. \end{cases} \tag{S.4}$$

Thus immediate acquisition is optimal exactly when $Q(x) \geq L$. If $B \geq P$, immediate acquisition is optimal as well.

Lemma **??** establishes Proposition **??**.

Proof. First establish the shape of Q. At a fixed capacity u, write $A = p'(u) > 0$, $c = -I'(u) > 0$ and $z = z(u)$. Then

$$g(x,u) = \begin{cases} g(z,u)(x/z)^\beta, & x < z, \\ Ax + p(u)c - AI(u), & x \geq z, \end{cases} \quad g(z,u) = \frac{AI(u)}{\beta - 1} + p(u)c.$$

Consequently

$$H_\alpha(x,u) := g(x,u) - \frac{xg_x(x,u)}{\alpha} = \begin{cases} (1-\beta/\alpha)g(x,u), & x < z, \\ (1-1/\alpha)Ax + p(u)c - AI(u), & x > z. \end{cases}$$

Both branches are strictly increasing. The right limit at z exceeds the positive left limit by $zp(u)c(\beta-1)/(\alpha I(u)) \geq 0$. Thus H_α is positive and strictly increasing, tending to zero as $x \downarrow 0$ and to infinity as $x \uparrow \infty$.

Differentiating (**??**) shows that Q is the same positive integral applied to H_α. It is therefore strictly increasing from zero to infinity, increases with κ, and satisfies $0 < Q < K$. The transition density smooths the branch changes; local differentiation is justified by the bounded coefficients on $[S, S+\kappa]$ and the integrable linear-growth bounds. In particular, K is locally C^2 and Q is locally C^1. These facts give a unique a with $Q(a) = L$.

For global verification, observe that

$$\frac{d}{dx}\frac{K(x) - L}{x^\alpha} = \alpha[L - Q(x)]x^{-\alpha-1}.$$

This ratio is maximized at a, so (**??**) majorizes the reward. Moreover, $K(a)-L = aK_x(a)/\alpha > 0$, which gives value matching and smooth pasting. Below a, $(\mathcal{L}-d)W = 0$. Above a, direct differentiation gives

$$(\mathcal{L}-d)(K-L) = d(L-Q) - \tfrac{1}{2}\sigma^2\alpha x Q_x \leq 0. \tag{S.5}$$

The candidate is nonnegative and C^1, with at most linear growth. Since $d > \mu$ and $d > 0$, discounted terminal values vanish. Localized Itô verification makes it an upper bound for every stopping rule; the first hitting time of a attains it, using $\mathbb{E}_x[e^{-d\tau_a}; \tau_a < \infty] = (x/a)^\alpha$ for $x < a$.

For the remaining case, $g(\cdot, u)$ is r-excessive: below z, $(\mathcal{L}-r)g = 0$; above z, $(r-\mathcal{L})g = (r-\mu)Ax + r[p(u)c - AI(u)] > 0$. The latter is already positive at z, where it equals $\frac{1}{2}\sigma^2\beta AI(u) + rp(u)c$. The downward derivative jump of g supplies an additional nonnegative loss term. Linear growth and $r > \mu$ justify the supermartingale statement. Equation (**??**) then makes K r-excessive as well: the state semigroup commutes with itself, and the inequality holds at every fixed capacity appearing in the integral. When $D = B - P \geq 0$, the candidate immediate value $D + K$ has no positive continuation gain: before conflict arrival its generator, including window closure and transition to direct value D upon conflict arrival, is

$$(\mathcal{L}-r)K - (r+\gamma_R)D - (\gamma_R+\lambda)K \leq 0.$$

After conflict arrival, delaying the nonnegative direct acquisition also cannot help. Immediate acquisition is therefore optimal. □

S.2.1 Quality thresholds and comparative statics

For $B = B_R(q)$ and $\kappa = \kappa_R(q, \rho_R)$, current surplus changes sign at

$$B_R(q^0) + K(x, S, \kappa_R(q^0, \rho_R)) = P_R.$$

Immediate acquisition begins at

$$B_R(q^*) + Q(x, S, \kappa_R(q^*, \rho_R)) = P_R. \tag{S.6}$$

If B_R is strictly increasing, κ_R is nondecreasing, and both left sides cross P_R, the thresholds are unique. Positive readiness and $Q < K$ imply $q^0 < q^*$. Existence follows, for example, from the linear specification $B_R = bq$ with $b > 0$. Below q^0, current surplus is negative; future acquisition can nevertheless have positive option value. This is not a rule for permanently excluding such parcels.

Several comparative statics follow directly. Higher acquisition cost raises q^*. More productive or more relevant readiness raises κ_R, increases Q, and lowers q^*. A larger window-closure hazard raises α without changing K, so

$$Q_{\gamma_R} = xK_x\alpha_{\gamma_R}/\alpha^2 > 0.$$

Faster depreciation reduces Q: write it as the expectation in (**??**) with $\mathcal{F}$ replaced by $\mathcal{F} - x\mathcal{F}_x/\alpha$. The capacity derivative of that expression is the positive H_α, while depreciation reduces the surviving increment. Hence faster depreciation raises q^*. These statements hold the other primitives fixed.

S.2.2 Interpreting the readiness contribution

For a fixed parcel with $L_R = P_R - B_R > 0$, let $W(x)$ denote its optimal acquisition value and a its acquisition threshold, so $Q(a) = L_R$. Value matching gives

$$W(a) = K(a) - L_R = \frac{aK_x(a)}{\alpha} > 0. \tag{S.7}$$

The additional value of retaining flexibility over the better of acquiring now and never acquiring is $W(x) - \max\{K(x) - L_R, 0\}$. It equals zero at a, although $aK_x(a)/\alpha$ is positive. The adjustment xK_x/α in Q therefore should not be described literally as the expected incremental gain from delaying acquisition.

S.2.3 The arrival-rate comparative static

Earlier conflict arrival raises readiness value. To see this for finite increments and depreciation, define

$$u(t,x) = e^{-rt}\mathbb{E}_x[\mathcal{F}(X_t, S + \kappa e^{-\delta t}) - \mathcal{F}(X_t, S)].$$

For any fixed increment, the payoff difference is r-excessive, since it is an integral of the r-excessive capacity derivative. Depreciation further reduces that increment. Thus $u(t,x)$ is nonincreasing in t. A larger λ makes T_C stochastically earlier, increasing $K = \mathbb{E}[u(T_C, x)]$.

For adaptive acquisition, a corresponding timing result is established for marginal increments when $\gamma_R \geq \delta$, and for finite increments when $\delta = 0$. Write $d_C = r + \lambda + \delta$ and $g = \mathcal{F}_S$. Differentiating $(d_C - \mathcal{L})\Omega = \lambda g$ gives

$$\Omega_\lambda = R_{d_C}(g - \Omega) \geq 0,$$

because $\Omega \leq [\lambda/(\lambda + \delta)]g \leq g$. If $\beta_C > 1$ and $\zeta < 0$ are the roots at discount rate d_C, the resolvent of any nonnegative source v is

$$R_{d_C}v(x) = \frac{2}{\sigma^2(\beta_C - \zeta)}\left[x^\zeta \int_0^x y^{-\zeta-1}v(y)\,dy + x^{\beta_C}\int_x^\infty y^{-\beta_C-1}v(y)\,dy\right].$$

Its logarithmic slope is at most β_C. Hence $x\Omega_{x\lambda} \leq \beta_C\Omega_\lambda$. When $\gamma_R \geq \delta$, the acquisition root satisfies $\alpha \geq \beta_C$, and

$$\left(\Omega - \frac{x\Omega_x}{\alpha}\right)_\lambda \geq \left(1 - \frac{\beta_C}{\alpha}\right)\Omega_\lambda + \frac{x\alpha_\lambda\Omega_x}{\alpha^2} > 0.$$

If $\delta = 0$, the finite-increment value is $K = \int_S^{S+\kappa}\Omega(x,u)\,du$, so integrating this inequality proves $Q_\lambda > 0$ for finite increments. No general finite-depreciation adaptive timing sign is claimed outside the proved conditions; no counterexample to that sign is asserted.

S.3 Acquisition policies: eligibility restrictions and future prices

S.3.1 Eligibility restrictions

Fix the parcel's quality, with $L_R = P_R - B_R > 0$ and $\kappa > 0$, and write $d = r + \lambda + \gamma_R$. Use a common filtration $\mathbb{F}$ under which the stakes retain their stated geometric Brownian dynamics. Eligibility and prices are adapted to $\mathbb{F}$, which is independent of both exit clocks; their survival probabilities are therefore included in e^{-dt}. Let $\mathcal{T}$ be the set of all $[0,\infty]$-valued $\mathbb{F}$-stopping

times, with $\tau = \infty$ denoting never acquiring at payoff zero. An eligibility rule, possibly based on an observed additionality threshold, restricts this set to $\mathcal{T}_E \subseteq \mathcal{T}$ containing never acquiring. Define

$$W(x) = \sup_{\tau \in \mathcal{T}} \mathbb{E}_x[\mathcal{R}(\tau)], \qquad W_E(x) = \sup_{\tau \in \mathcal{T}_E} \mathbb{E}_x[\mathcal{R}(\tau)], \tag{S.8}$$
$$\mathcal{R}(\tau) = \begin{cases} e^{-d\tau}[K(X_\tau) - L_R], & \tau < \infty, \\ 0, & \tau = \infty. \end{cases}$$

Only admissible acquisition dates change: information, payoffs, and exit risks are identical.

Lemma S.3.1 (Eligibility and the value of protection)**.** *Under the assumptions of Proposition* **??**, $0 \le W_E(x) \le W(x)$. *Let a be the threshold of Lemma* **??**, *so $Q(a) = L_R$. If $x > a$ and there is a fixed $\varepsilon > 0$ such that every $\tau \in \mathcal{T}_E$ satisfies $\tau \ge \varepsilon$ almost surely on $\{\tau < \infty\}$, then*

$$W_E(x) \le \mathbb{E}_x[e^{-d\varepsilon} W(X_\varepsilon)] < W(x) = K(x) - L_R. \tag{S.9}$$

The result requires neither bounded stopping times nor attainment of the constrained supremum.

Proof of Lemma **??**. Set inclusion and the option never to acquire give the weak inequalities. The verification of Lemma **??** shows that $M_t = e^{-dt} W(X_t)$ is a nonnegative supermartingale and $W \ge K - L_R$. For every eligible $\tau \ge \varepsilon$, optional sampling at $\tau \wedge n$ and Fatou's lemma give

$$\mathbb{E}_x[\mathcal{R}(\tau)] \le \mathbb{E}_x[M_\tau \mathbf{1}_{\{\tau < \infty\}}] \le \mathbb{E}_x[M_\varepsilon], \tag{S.10}$$

with the integrand set to zero when $\tau = \infty$.

For strictness, choose $a < u < x < v < \infty$ and let ρ be the first exit from (u, v). Equation (**??**) gives the continuous loss rate

$$c(y) = -(\mathcal{L} - d)W(y) = d[Q(y) - L_R] + \tfrac{1}{2}\sigma^2 \alpha y Q_x(y) > 0, \qquad y > a. \tag{S.11}$$

Hence $c_0 = \min_{y \in [u,v]} c(y) > 0$, and path continuity gives $\rho > 0$ almost surely. Itô's formula stopped inside this compact interval, followed by optional sampling, yields

$$\begin{aligned} \mathbb{E}_x[M_\varepsilon] &\le \mathbb{E}_x[M_{\varepsilon \wedge \rho}] \\ &= W(x) - \mathbb{E}_x\left[\int_0^{\varepsilon \wedge \rho} e^{-ds} c(X_s)\, ds\right] \\ &\le W(x) - c_0 \mathbb{E}_x\left[\int_0^{\varepsilon \wedge \rho} e^{-ds}\, ds\right] < W(x). \end{aligned}$$

This fixed positive gap and (**??**) prove the strict comparison after taking the supremum over eligible policies. □

Excluding immediate acquisition alone need not cause a strict loss if eligible dates can approach zero. If a deterministic release date ε is the sole restriction, waiting until release and then following the optimal adaptive policy attains the lemma's upper bound; acquisition need not occur at release.

S.3.2 Higher future acquisition prices

Consider two nonnegative optional price processes on the same filtered space, satisfying, almost surely simultaneously for all $t \geq 0$,

$$P^{(1)}_{R,0} = P^{(0)}_{R,0} = P_R, \qquad P^{(1)}_{R,t} \geq P^{(0)}_{R,t}. \tag{S.12}$$

Both problems use the same information, admissible set $\mathcal{T}$, stakes, readiness technology, and direct benefit. The opportunity still ends at either independent exit event; for example, future prices above B_R maintain the readiness-dependent interpretation. No post-exit acquisition opportunity is added. Define

$$W^{(j)}(x) = \sup_{\tau \in \mathcal{T}} \mathbb{E}_x[\mathcal{R}_j(\tau)], \qquad \mathcal{R}_j(\tau) = \begin{cases} e^{-d\tau}[B_R + K(X_\tau) - P^{(j)}_{R,\tau}], & \tau < \infty, \\ 0, & \tau = \infty. \end{cases} \tag{S.13}$$

Additional observed price variables are conditioned on at the common initial state and suppressed in the notation.

Lemma S.3.2 (Higher future prices and immediate acquisition)**.** *Suppose* (**??**) *holds and both immediate acquisition and never acquiring are admissible. Then*

$$B_R + K(x) - P_R \leq W^{(1)}(x) \leq W^{(0)}(x). \tag{S.14}$$

Thus immediate acquisition remains optimal whenever it was optimal under the lower future prices. Holding today's price fixed, higher future prices weakly expand the immediate-acquisition region.

Proof of Lemma **??**. Since $0 \leq F(y; I) \leq y$ and $p \leq 1$, (**??**) gives

$$0 \leq K(y) \leq c_K y, \qquad c_K = \frac{\lambda}{r + \lambda - \mu}. \tag{S.15}$$

The process $e^{-dt}X_t$ is a nonnegative supermartingale. Optional sampling and Fatou's lemma therefore give $\mathbb{E}_x[e^{-d\tau}K(X_\tau)\mathbf{1}_{\{\tau<\infty\}}] \leq c_K x$ for every $\tau \in \mathcal{T}$. Thus $\mathbb{E}_x[\mathcal{R}_j(\tau)^+] \leq B_R + c_K x$; each expected reward is well defined, allowing value $-\infty$ when its negative part is not integrable. Such policies cannot improve on never acquiring.

The simultaneous pathwise price ordering gives $\mathcal{R}_1(\tau) \leq \mathcal{R}_0(\tau)$ for every common policy, hence $W^{(1)} \leq W^{(0)}$. Immediate acquisition gives the same payoff $J(x) = B_R + K(x) - P_R$ in both problems, so $J \leq W^{(1)}$. If $W^{(0)} = J$, these inequalities force $W^{(1)} = J$ as well. □

This proves Proposition **??** for stochastic prices and unbounded stopping times. It does not require a constant acquisition threshold or imply a strict ordering of acquisition dates when both policies wait.

S.4 The capacity threshold for the value of readiness: proof of Proposition ??

Proposition **??** concerns a global capacity crossing for the marginal value of readiness. The proof uses both branches of the intervention option.

Fix $x > 0$, $\sigma > 0$, $\lambda > 0$, $\delta \geq 0$, and $r > \max\{\mu, 0\}$. Let $F(x; I; \sigma)$ be the perpetual intervention option, including its immediate-intervention branch. Define

$$G(x, I; \sigma) = \lambda \int_0^\infty e^{-(r+\lambda+\delta)t} \mathbb{E}_x^\sigma[F(X_t; I; \sigma)]\, dt. \tag{S.16}$$

The marginal value of a readiness increment, $\Omega = \partial K/\partial \kappa|_{\kappa=0}$, is

$$\Omega(x, S; \sigma) = p'(S)G(x, I(S); \sigma) + p(S)I'(S)G_I(x, I(S); \sigma). \tag{S.17}$$

Depreciation enters the marginal increment through the discount adjustment δ. Volatility derivatives below hold x, S fixed for Ω and x, I fixed for G.

Statement. The following restates Proposition **??**. Suppose p and I are sufficiently smooth on $[0, \infty)$ and satisfy, *globally*,

$$\begin{aligned} &p(0) = 0, \qquad p' > 0, \qquad p'' \leq 0, \qquad 0 \leq p \leq 1, \\ &I(0) > 0, \qquad \chi_I(S) := -I'(S)/I(S) > 0, \qquad \chi_I'(S) \geq 0. \end{aligned} \tag{S.18}$$

There is exactly one $S^* \in (0, \infty)$ such that

$$\Omega_\sigma(x, S; \sigma) \begin{cases} > 0, & 0 \leq S < S^*, \\ = 0, & S = S^*, \\ < 0, & S > S^*. \end{cases} \tag{S.19}$$

The result holds at any current stakes, including stakes at or above the intervention threshold. The statement concerns the marginal readiness coefficient, not a finite depreciating increment or the optimal acquisition date.

S.4.1 The full resolvent and its volatility derivative

Put $\lambda_\delta = \lambda + \delta$, $d_C = r + \lambda_\delta$, and $c = \lambda/\lambda_\delta$. Let $\beta > 1$ solve

$$\tfrac{1}{2}\sigma^2\beta(\beta - 1) + \mu\beta = r,$$

and let $\beta_C > \beta$ and $\zeta < 0$ be the two roots of

$$\tfrac{1}{2}\sigma^2 s(s - 1) + \mu s = d_C.$$

For $z = \beta I/(\beta - 1)$ and $u = x/z$, solving the resolvent equation $(d_C - \mathcal{L})G = \lambda F$ gives

$$G(x, I; \sigma) = \begin{cases} \dfrac{cI}{\beta - 1} u^\beta - C_- I u^{\beta_C}, & u \leq 1, \\ \lambda\left(\dfrac{x}{d_C - \mu} - \dfrac{I}{d_C}\right) + C_+ I u^\zeta, & u \geq 1, \end{cases} \tag{S.20}$$

where

$$C_- = \frac{c\beta(\beta - \zeta)}{\beta_C(\beta_C - 1)(\beta_C - \zeta)}, \qquad C_+ = \frac{c\beta(\beta_C - \beta)}{\zeta(\zeta - 1)(\beta_C - \zeta)}. \tag{S.21}$$

Both coefficients are positive. The lower branch vanishes at zero; the upper branch has at most linear growth. Matching value and first derivative at z identifies the coefficients. The differential equation also makes G_{xx} continuous there, since F is continuous.

The root derivatives are

$$\beta_\sigma = -\frac{2\beta(\beta-1)}{\sigma(2\beta-\beta_C-\zeta)}, \qquad \beta_{C,\sigma} = -\frac{2\beta_C(\beta_C-1)}{\sigma(\beta_C-\zeta)}, \tag{S.22}$$

$$\zeta_\sigma = \frac{2\zeta(\zeta-1)}{\sigma(\beta_C-\zeta)} > 0, \qquad \frac{z_\sigma}{z} = -\frac{\beta_\sigma}{\beta(\beta-1)}. \tag{S.23}$$

The threshold's volatility dependence must be included when differentiating equation (**??**).

Write $H = G_\sigma$. For the lower branch define

$$t = \log(z/x) \geq 0, \qquad k = \beta_C - \beta > 0, \qquad w = \frac{\beta-\zeta}{\beta_C-\zeta} \in (1/2, 1), \qquad C = \frac{c(-\beta_\sigma)}{\beta-1} > 0.$$

The bound on w follows from $\beta_C > \beta$ and $2\beta - \beta_C - \zeta > 0$. Differentiation at fixed x, I, using $t_\sigma = z_\sigma/z$, gives

$$\frac{H}{I} = Ce^{-\beta t}\left[t + e^{-kt}(B - d_0 t)\right], \qquad d_0 = w(2w-1), \qquad B = \frac{4w(1-w)}{\beta_C-\zeta}. \tag{S.24}$$

For verification, the two coefficient identities in this differentiation are

$$\frac{-C_-\beta_{C,\sigma}}{C} = d_0, \qquad \frac{C_-\beta_C(z_\sigma/z) - C_{-,\sigma}}{C} = B.$$

Here $0 < d_0 < 1$ and $B > 0$.

On the upper branch,

$$\frac{H}{I} = u^\zeta\left[E_0 + C_+\zeta_\sigma \log u\right], \qquad E_0 = C_{+,\sigma} - C_+\zeta z_\sigma/z. \tag{S.25}$$

At $u = 1$, equation (**??**) gives $E_0 = CB > 0$. Equations (**??**)–(**??**) therefore establish $H > 0$ everywhere. In particular, with $p(0) = 0$,

$$\Omega_\sigma(x, 0; \sigma) = p'(0)H(x, I(0); \sigma) > 0.$$

This low-capacity result reflects the value of access: at zero capacity, the first marginal increment makes intervention feasible rather than reducing the cost of an intervention already available.

S.4.2 Global log-concavity and the capacity crossing

We next show that H/I is strictly log-concave as a function of $\log(x/I)$. For the lower branch set

$$v = kt, \qquad \varepsilon_0 = kB = 4w(1-w)^2, \qquad h(v) = v + e^{-v}(\varepsilon_0 - d_0 v).$$

The bracket in equation (**??**) equals $h(v)/k$. Moreover,

$$h(v) = v(1 - d_0 e^{-v}) + \varepsilon_0 e^{-v} > 0.$$

Direct calculation gives

$$h'^2 - hh'' = 1 - e^{-v}[2d_0 + 2\varepsilon_0 + \varepsilon_0 v - d_0 v^2] + d_0^2 e^{-2v}. \tag{S.26}$$

The parameter identity

$$(1-d_0)^2 - 2\varepsilon_0 = (1-w)^2(2w-1)^2 > 0 \tag{S.27}$$

implies $\varepsilon_0 < (1-d_0)^2/2$. The right-hand side of equation (**??**) decreases with ε_0. Replace it by the larger value $(1-d_0)^2/2$ and multiply the resulting lower bound by e^v. The result is

$$E(v) = e^v + d_0^2 e^{-v} - 2d_0 + d_0 v^2 - \tfrac{1}{2}(1-d_0)^2(v+2).$$

It satisfies

$$E(0) = 0, \qquad E'(0) = \tfrac{1}{2}(1-d_0)(1+3d_0) > 0, \qquad E''(v) = e^v + d_0^2 e^{-v} + 2d_0 > 0.$$

Hence $E(v) > 0$ for $v > 0$. At $v = 0$, the actual determinant is strictly positive by equation (**??**). Thus $h'^2 - hh'' > 0$ throughout the lower branch. Taking its logarithm proves strict concavity; the exponential factor in equation (**??**) adds only a linear term, and v is affine in $\log(x/I)$.

On the upper branch, the logarithm of equation (**??**) is a linear function of $\log u$ plus $\log(E_0 + C_+\zeta_\sigma \log u)$, which is strictly concave. The value and first derivative of H match at z. To see this, differentiate the matching conditions for G and G_x with respect to σ and use the continuity of G_x and G_{xx}. The logarithmic slope consequently has no jump at the boundary. Strict log-concavity therefore holds globally.

Homogeneity allows us to write $H(x, I) = I\widehat{H}(y)$, $y = x/I$. Consequently,

$$m_H(y) := \frac{IH_I}{H} = 1 - \frac{\partial \log \widehat{H}(y)}{\partial \log y} \tag{S.28}$$

is strictly increasing. The tail expressions above give

$$\lim_{y \downarrow 0} m_H(y) = 1 - \beta < 0, \qquad \lim_{y \to \infty} m_H(y) = 1 - \zeta > 0. \tag{S.29}$$

Differentiating equation (**??**) yields

$$\frac{\Omega_\sigma}{H} = p' - p\chi_I m_H(x/I). \tag{S.30}$$

For $S > 0$, its sign is therefore the sign of $p'/(p\chi_I) - m_H(x/I)$.

The first term strictly decreases, because

$$\frac{d}{dS}\frac{p'}{p\chi_I} = \frac{p''p\chi_I - (p')^2\chi_I - p'p\chi_I'}{p^2\chi_I^2} < 0.$$

It tends to infinity as $S \downarrow 0$. Bounded increasing concave access implies $p' \to 0$ and $p \to p_\infty > 0$, while $\chi_I(S) \geq \chi_I(0) > 0$, so it tends to zero as $S \to \infty$. Also $I(S) \leq I(0)e^{-\chi_I(0)S}$, implying $x/I(S) \to \infty$. Thus the second term strictly increases toward $1 - \zeta > 0$. Their difference is continuous and strictly decreasing from positive to negative. This proves the unique crossing in equation (**??**).

S.4.3 What the theorem establishes, and what timing still requires

The global assumptions matter. If $p(0) > 0$, the same argument gives at most one crossing; existence requires a positive initial volatility response. If feasible capacity has a finite upper bound, the crossing need not lie within that range. The theorem does not establish a global crossing for large, depreciating readiness increments.

Nor does the crossing for readiness value alone determine acquisition timing. The full derivative of $Q = K - xK_x/\alpha$ is given in equation (**??**); its marginal counterpart follows

by replacing K with Ω. Online Appendix **??** establishes conditions under which a positive readiness-value response carries over to immediate protection.

S.5 Fragile acquisition windows: proofs of Proposition ?? and Corollary ??

Local assumptions and conclusions. Maintain the model assumptions in Appendix **??**, including $r > \max\{\mu, 0\}$, $\lambda > 0$ and $\delta \geq 0$. Fix $x > 0$ and a reference volatility $\sigma_0 > 0$, and write $\kappa(q) = \kappa_R(q, \rho_R)$ with ρ_R fixed. All primitives other than the parameter being varied are held fixed; in particular, K is independent of γ_R. Assume B_R, κ are continuously differentiable near the relevant cutoff, with $B_R' > 0$ and $\kappa' \geq 0$, and that $K, K_x, K_\sigma, K_\kappa, K_{x\sigma}, K_{x\kappa}$ are continuous locally for $\kappa > 0$ and $\sigma > 0$. At $S = 0$, continuity is relative to $S \geq 0$. The cutoffs solve

$$\begin{aligned} B_R(q^0) + K(x, S, \kappa(q^0); \sigma) &= P_R, \\ B_R(q^*) + Q(x, S, \kappa(q^*); \sigma, \gamma_R) &= P_R. \end{aligned} \tag{S.31}$$

Proposition **??** assumes that q^0 is interior at σ_0, that $\kappa(q^0) > 0$, and that $K_\sigma(x, S, \kappa(q^0); \sigma_0) > 0$. In addition to its stated sign conclusion, we show that $q^* \downarrow q^0$ as $\gamma_R \to \infty$ and

$$\left.\frac{\partial q^*}{\partial \sigma}\right|_{\sigma=\sigma_0} \longrightarrow -\frac{K_\sigma(x, S, \kappa(q^0); \sigma_0)}{B_R'(q^0) + K_\kappa(x, S, \kappa(q^0); \sigma_0)\kappa'(q^0)} < 0. \tag{S.32}$$

No global sign or single-crossing condition on K_σ is assumed.

S.5.1 Fragile-window comparative static

At the positive root in (**??**), put $D_\alpha = \sigma^2\alpha + \mu - \sigma^2/2 > 0$. Implicit differentiation gives

$$\begin{aligned} \alpha_\sigma &= -\frac{\sigma\alpha(\alpha-1)}{D_\alpha}, & \alpha_{\gamma_R} &= \frac{1}{D_\alpha}, \\ Q_\sigma &= K_\sigma - \frac{x}{\alpha}\left(K_{x\sigma} + \frac{\sigma(\alpha-1)}{D_\alpha}K_x\right). \end{aligned} \tag{S.33}$$

Thus an explicit sufficient condition at a finite closure rate is

$$K_\sigma > \frac{x}{\alpha}\left(|K_{x\sigma}| + \frac{\sigma(\alpha-1)}{D_\alpha}K_x\right) \quad \Longrightarrow \quad Q_\sigma > 0. \tag{S.34}$$

The coefficient $\sigma(\alpha-1)/D_\alpha$ need not be bounded above by $1/\sigma$ when $\mu + \sigma^2/2 < 0$.

Proof of Proposition **??**. As $\gamma_R \to \infty$, $\alpha \sim \sqrt{2\gamma_R}/\sigma$ and $\sigma(\alpha-1)/D_\alpha \to 1/\sigma$, uniformly for σ in compact subsets of $(0, \infty)$. Equation (**??**) and $Q_\kappa = K_\kappa - xK_{x\kappa}/\alpha$ therefore imply

$$Q \to K, \qquad Q_\sigma \to K_\sigma, \qquad Q_\kappa \to K_\kappa \quad \text{locally uniformly.} \tag{S.35}$$

The limiting current-surplus equation has positive quality derivative $B_R' + K_\kappa\kappa' > 0$, since $K_\kappa \geq 0$. Choose a small quality interval around q^0 with opposite endpoint signs. For σ near σ_0 and sufficiently large γ_R, those signs persist uniformly, while the acquisition equation has derivative $B_R' + Q_\kappa\kappa' > 0$. It therefore has a unique nearby root $q^* \to q^0$. The implicit-function theorem and differentiation of (**??**), as in (**??**), now give (**??**). The strictly negative limit implies $q_\sigma^* < 0$ for every sufficiently large finite γ_R. Finally, $Q < K$ and $Q_{\gamma_R} = xK_x/(\alpha^2 D_\alpha) > 0$ imply $q^* > q^0$ and $q_{\gamma_R}^* < 0$, so convergence is from above. □

Rate of convergence. At fixed σ_0, the same cutoff equation gives

$$q^* - q^0 = \frac{xK_x(x, S, \kappa(q^0); \sigma_0)}{\alpha\{B_R'(q^0) + K_\kappa(x, S, \kappa(q^0); \sigma_0)\kappa'(q^0)\}} + o(\alpha^{-1}). \tag{S.36}$$

The positive limiting quality derivative first gives $q^* - q^0 = O(\alpha^{-1})$; a first-order expansion and continuity of K_x give the displayed coefficient.

S.5.2 Finite readiness increments at low capacity

Use the option value $F(y; I, \sigma)$, root $\beta > 1$ and threshold $z = \beta I/(\beta - 1)$ from Appendix **??**. Implicit differentiation gives $\beta_\sigma < 0$. Below z, $F_\sigma = \beta_\sigma F \log(y/z) > 0$; above z it is zero. The option is convex, and its value and first state derivative match at z, so there is no extra threshold term in the derivatives below.

For fixed $t > 0$ and $I > 0$, Gaussian integration by parts (**?**, Lemma 1) gives

$$\frac{\partial}{\partial\sigma}\mathbb{E}_x[F(X_t^\sigma; I, \sigma)] = \mathbb{E}_x[F_\sigma(X_t^\sigma; I, \sigma)] + \sigma t\, \mathbb{E}_x[(X_t^\sigma)^2 F_{yy}(X_t^\sigma; I, \sigma)]. \tag{S.37}$$

Indeed, use $X_t^\sigma \overset{d}{=} x\exp[(\mu - \sigma^2/2)t + \sigma\sqrt{t}\, Z]$ and $\mathbb{E}[Zh(Z)] = \mathbb{E}[h'(Z)]$ with $Z \sim N(0,1)$ and $h(Z) = X_t^\sigma F_y(X_t^\sigma; I, \sigma)$. Smooth pasting, $0 \le F_y \le 1$ and lognormal moments justify this step. The option value also gives

$$0 \le F_\sigma \le \frac{|\beta_\sigma| I}{e\beta(\beta - 1)}, \qquad 0 \le y^2 F_{yy} \le \beta I \quad \text{almost everywhere.} \tag{S.38}$$

On compact positive-volatility and finite-capacity ranges, I and the root coefficients are bounded, so (**??**) is bounded by $C(1 + t)$, whose discounted integral is finite. Since X_t^σ has no atom at z for $t > 0$, these bounds justify differentiation and continuity of K_σ under the exponential-arrival expectation in (**??**), including the one-sided limit at $S = 0$.

Proof of Corollary **??**. Fix any finite $\kappa > 0$ and set $u_t = \kappa e^{-\delta t}$. When $S = 0$, $p(0) = 0$ eliminates the baseline term in (**??**), giving

$$K_\sigma(x, 0, \kappa; \sigma) = \int_0^\infty \lambda e^{-(r+\lambda)t} p(u_t) \frac{\partial}{\partial\sigma}\mathbb{E}_x[F(X_t^\sigma; I(u_t), \sigma)]\, dt > 0. \tag{S.39}$$

For every finite $t > 0$, $u_t > 0$ and $p(u_t) > 0$. Both terms in (**??**) are nonnegative; the first is strictly positive because the lognormal distribution has positive probability below z, where $F_\sigma > 0$. This proves the strict inequality for any finite increment and any fixed $\delta \ge 0$.

Let q_0^0 be the interior current-surplus cutoff at $(S, \sigma) = (0, \sigma_0)$, with $\kappa(q_0^0) > 0$. By continuity, choose a compact neighborhood of this point, relative to $S \ge 0$, on which $\kappa(q) > 0$, K_σ and B_R' have positive lower bounds, and fixed quality endpoints bracket the current-surplus root. The convergences in (**??**) are uniform on this neighborhood. Thus the preceding root argument gives a single finite $\bar{\gamma}_R$ for all sufficiently small $0 \le S \le \bar{S}$: the acquisition root stays in the bracket and has $Q_\sigma > 0$. Equation (**??**) then gives $q_\sigma^* < 0$ throughout this range. No derivative with respect to S is required. □

S.5.3 Illustrative finite-increment examples

Use

$$r = .05, \quad \mu = .02, \quad \lambda = .05, \quad \gamma_R = .05, \quad \delta = 0, \quad p(S) = 1 - e^{-S}, \quad I(S) = e^{-.5S}, \quad \kappa = .1.$$

Current stakes are $x = .1$. For each capacity case, choose a fixed direct shortfall L that places the acquisition trigger at .1 when $\sigma = .30$, then hold L fixed as volatility varies. The resulting triggers are:

Initial capacity	Direct shortfall L	$\sigma = .29$	$\sigma = .30$	$\sigma = .31$
$S = .1$	0.0006085015262	.102715991	.100000000	.097481438
$S = 2.8$	0.0002426691792	.098778173	.100000000	.101303776

For the lower-capacity case, greater uncertainty expands the immediate-acquisition region. At the same current state .1, the decision changes from waiting to acquisition. For the higher-capacity case, the direction reverses. These examples use the option value at C, a finite increment, and an adaptive acquisition policy.

The direct shortfalls differ between the two rows because the examples are aligned at the same initial boundary. They establish that both comparative-static directions occur in the model. They do not establish a universal capacity threshold for acquisition timing. A change in the acquisition trigger also does not, by itself, compare expected calendar waiting times across stochastic processes with different volatilities.

S.6 A protection date fixed in advance

Here the agency commits to a deterministic acquisition date, cancelled by acquisition-window closure or arrival of the later conflict; it does not revise the date as stakes change. Use the marginal approximation $K \approx \kappa\Omega$, with Ω as in (**??**), and set $L = P_R - B_R > 0$, $h = \gamma_R + \lambda$, $a_0 = r + h$, $D_\Omega = (r - \mathcal{L})\Omega > 0$, and $J = h\Omega + D_\Omega$. Within this approximation the planned-date payoff and its derivative are

$$V(t) = e^{-a_0 t}\{\kappa \mathbb{E}_x \Omega(X_t) - L\},$$
$$V'(t) = e^{-a_0 t}\{a_0 L - \kappa \mathbb{E}_x J(X_t)\}.$$

The timing payoff need not have a single peak. For $p(S) = 1 - e^{-S}$, $I(S) = e^{-S/2}$, $r = .05$, $\mu = .02$, $\sigma = .30$, $\lambda = \gamma_R = .05$, $\delta = 0$, $S = 2.8$, $x = z(S) = .9169588273$, and $L/\kappa = .0681$, truncated-moment evaluation gives $V(0)/\kappa = .0002165243$ and $V'(0)/\kappa = -.0010290982$, but $V(15.3492)/\kappa \approx .000416 > V(0)/\kappa$. Immediate acquisition is therefore a strict local maximum dominated by a later date.

Comparing $V(0)$ with all $V(t)$ gives the valid global condition $\kappa\Omega(x) \geq M_{\text{full}} L$, where

$$M_{\text{full}} = \Omega(x) \sup_{t>0} \frac{1 - e^{-a_0 t}}{\Omega(x) - e^{-a_0 t}\mathbb{E}_x \Omega(X_t)}, \qquad 1 \leq M_{\text{full}} < 1 + r/h.$$

Strict r-excessivity and the limits at zero and infinity give these bounds. The lower bound is strict exactly when $\mathbb{E}_x \Omega(X_t) > \Omega(x)$ for some t; $\mu > 0$ suffices. The adaptive rule does not rely on planned-date unimodality.

S.7 Strategic extension: contests, deterrence, and the landscape

S.7.1 A developer threshold for contest formation

Let $Z \sim G$, with density g, be the private gain from securing R relative to the developer's fallback. With contest cost $c_D(S,\theta)$ and fixed win probability $\pi_D \in (0,1)$, contesting yields $\pi_D Z - c_D(S,\theta)$. The developer therefore contests iff $Z \geq J(S,\theta)$, where

$$J(S,\theta) = \frac{c_D(S,\theta)}{\pi_D}, \qquad J_S > 0, \qquad J_\theta < 0.$$

Here S is conservation capacity and θ is developer strength. More general differences in the developer's private fallback after deterrence and defeat can be absorbed into the reduced-form hurdle J. The probability of no contest and its derivatives are

$$n(S,\theta) = G(J(S,\theta)), \qquad n_S = g(J)J_S \geq 0, \qquad n_\theta = g(J)J_\theta \leq 0,$$

with strict inequalities wherever $g(J) > 0$.

For bounded prizes $Z \leq \overline{Z}$ and the hurdle $J(S,\theta) = J_0 e^{\alpha_D S}/\theta$, with $J_0, \alpha_D, \theta > 0$,

$$S_{\text{det}}(\theta) = \max\{0, \alpha_D^{-1} \log(\theta \overline{Z}/J_0)\}.$$

With a continuous prize distribution, no contest occurs almost surely for $S \geq S_{\text{det}}(\theta)$.

At fixed θ, write $c(S) = 1 - n(S,\theta)$. Causal accounting restricts contests to projects that would have developed R without protection, so $0 \leq c(S) \leq \xi$, and holds π_D fixed. At the boundary $\pi_D = 0$, positive contest costs eliminate contests; savings in contest resources lie outside the ecological accounting.

S.7.2 Focal and landscape accounting

The four outcomes have probabilities $c(1-\pi_D)$ (successful defense), $c\pi_D$ (successful developer contest), $\xi - c$ (deterred development), and $1 - \xi$ (no counterfactual development). Only the second produces focal damage, giving equation (**??**) and Proposition **??**:

$$\frac{\partial A_R^{\text{obs}}}{\partial S} = (1-\pi_D)c'(S)\Lambda_R \leq 0, \qquad \frac{\partial A_R^{\text{di}}}{\partial S} = -\pi_D c'(S)\Lambda_R \geq 0.$$

Subtracting protected landscape damage $c\pi_D\Lambda_R + c(1-\pi_D)L^C + (\xi - c)L^D$ from $\xi\Lambda_R$ gives equation (**??**). Holding π_D, L^C, L^D fixed and using $n = 1 - c$,

$$\frac{\partial A_R^{\text{land}}}{\partial n} = \pi_D(\Lambda_R - L^C) - (L^D - L^C),$$

which gives the reversal condition (**??**). With $L^C = L^D = L$, it reduces to $\pi_D(\Lambda_R - L)$.

S.7.3 Value-first targeting and landscape outcomes

Proof of Proposition **??**. Every feasible relocation destination has total loss no greater than Λ_R. Averaging over abandonment and relocation therefore gives $L^C, L^D \leq \Lambda_R$. The probability weights in equation (**??**) are nonnegative, so $A_R^{\text{land}}(S) \geq 0$, strictly whenever at least one blocking outcome has positive probability and its expected fallback damage is strictly below Λ_R. □

S.8 Figure definitions and normalization

Figure **??** is a schematic summary rather than a plot. Panel (a) orders q^0 and q^* as in Proposition **??**, which requires positive readiness so that $0 < Q < K$; the arrows record the comparative statics of Section **??**. Panel (b) marks S^*, where the volatility response of the marginal value of readiness changes sign; it holds under the conditions of Proposition **??**, and the corresponding change in the timing decision is the subject of Section **??**.

Figure **??** considers the readiness-dependent case $B_R < P_R$, with a positive readiness increment and current stakes $x > a$. All model primitives are held fixed. The earliest eligible protection date τ is deterministic, and the only added restriction is that protection cannot occur before τ; afterward the agency can follow the ordinary adaptive acquisition policy. The constrained value is

$$W_E(x;\tau) = \mathbb{E}_x\left[e^{-(r+\lambda+\gamma_R)\tau}W(X_\tau)\right].$$

Waiting until τ and then following the optimal policy attains the upper bound in Lemma **??**. This value starts at $W(x)$, is nonincreasing in τ, and is strictly below $W(x)$ for every $\tau > 0$ when $x > a$. The agency's value and near-horizon additionality are separately normalized, so only their timing patterns are compared. For display, the schematic blue curve $e^{-\tau/6}$ illustrates the declining shape of normalized constrained value; it is not a numerical evaluation of the displayed expectation for a specified structural parameterization. The illustrative additionality index is $A(\tau) = 1 - e^{-\tau/3.5}$, with eligibility threshold $\bar{A} = 0.64$, so $\tau_A = -3.5\log(0.36) \approx 3.58$. Figure **??** assumes $L^C = L^D = L < \Lambda_R$ and normalizes by $\xi(\Lambda_R - L)$. The contest-visible component is $c(S)(1-\pi_D)(\Lambda_R - L)$, total landscape benefit is $[\xi - \pi_D c(S)](\Lambda_R - L)$, and their difference is the deterrence component $[\xi - c(S)](\Lambda_R - L)$. The schematic sets $c(S)/\xi = 1 - (S/S_{\text{det}})^{1.8}$ with $S_0 = 0$ and $\pi_D = 0.32$, so all at-risk projects contest at S_0 and capacity reduces this probability to zero at S_{det}. These are illustrative paths, not calibrated estimates.